Mutual Visual HIL Model Comparison for Firefly-Inspired Synchronisation

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Abstract

This report analyses a formal mutual hardware-in-the-loop (HIL) comparison between EAPF Consensus and Kuramoto for firefly-inspired visual synchronisation. The experiment uses one browser-rendered virtual agent and one Raspberry Pi visual agent in a bidirectional loop: the Pi camera observes the virtual flash, the Pi oscillator flashes its GPIO LED, and Pi flash events are posted back to the frontend. The formal data comprise 18 completed trials: two models, three Pi initial frequencies, and three repeats with controlled random initial phases. Under this camera-mediated event-based mutual HIL condition, EAPF Consensus achieves lower final-window frequency error and much lower frequency variability than Kuramoto with locked $K = 5.0$. An appendix Kuramoto gain sensitivity sweep over $K \in \{2.5, 3.0, 3.5, 4.0, 4.5\}$ finds that $K = 2.5$ improves behaviour but does not fully resolve the oscillatory response. EAPF remains the model selected for the next 2-virtual + 1-Pi mutual HIL stage.

1 Introduction

The broader project investigates decentralised visual flashing as a mechanism for coordinating multi-drone behaviour. Earlier stages established simulation models, camera flash detection, GPIO LED output, a browser-based visual leader, a fixed-leader visual HIL comparison, and a simulation-based multi-neighbour model selection. The fixed-leader setting is an important but simpler one-way tracking problem: the virtual display acts as a stable external reference and the Pi follows it without the leader reacting to the follower.

This report extends the evaluation to mutual visual HIL. In this mode, the virtual agent is no longer a fixed leader. Instead, the Pi observes the virtual flash through the camera, while its own flashes are sent back to the frontend over the HTTP API. This creates a closed-loop bidirectional coupling problem. The purpose of this report is to evaluate which of the two model candidates, EAPF Consensus and Kuramoto, is more robust in this mutual event-based visual loop.

This report does not introduce new fixed-leader experiments and does not claim universal model superiority. It asks a narrower question: based on the formal mutual visual HIL data currently available, which model is the better engineering candidate for the next 2-virtual + 1-Pi mixed-reality multi-agent stage?

2 Experimental Objective

The objective was to compare EAPF Consensus and Kuramoto under matched 1-virtual + 1-Pi mutual HIL conditions. The virtual reference frequency was 2.0 Hz. The Pi oscillator was initialised at 1.2 Hz, 1.5 Hz, or 2.5 Hz to test convergence across below-reference, moderate-mismatch, and above-reference conditions. Each model-condition pair was repeated three times with controlled random initial phases.

The primary outcome is mutual frequency agreement in the final 5 s of a 30 s trial. Secondary outcomes include virtual flash detection reliability, flash counts, runtime diagnostics, and variability across repeats.

3 Method

3.1 Mutual HIL setup

The 1-virtual + 1-Pi mutual HIL loop contains one browser-rendered virtual agent and one real Pi agent. The frontend displays the virtual flash as a high-contrast target. The Pi camera detects the virtual flash, updates its local oscillator, and drives the GPIO LED. When the Pi oscillator fires, the runner posts a timestamped event to the frontend API. The frontend server then updates the virtual agent from those Pi flash events.

This bidirectional loop is substantially different from fixed-leader visual HIL. In fixed-leader mode the visual target is an exogenous, stable reference: the leader flashes at a fixed frequency and does not react to the follower. The follower's coupling corrections affect only its own phase and frequency. In mutual HIL, both agents influence each other, so unstable feedback or event-timing sensitivity can appear even if a model worked well in the one-way tracking case.

3.2 Continuous phase coupling versus binary flash events

An important architectural distinction affects the model comparison. Kuramoto is naturally a continuous phase-coupled model: in its standard form the coupling term $\sin(\theta_j - \theta_i)$ uses continuously available neighbour phase. In pure simulation, that is a natural assumption.

In the implemented mutual visual HIL system, however, the camera stream is continuous in time but the binary flash detection signal mainly provides sparse event timestamps. Continuous neighbour phase is reconstructed from those flash timestamps—estimated as $\hat{\theta} = 2\pi(t - t_{\text{last}})/T_{\text{est}}$, where T_{est} is the median of recent inter-flash intervals. The resulting interaction is therefore an event-derived approximation to continuous phase coupling.

EAPF Consensus was designed from the start for event-based phase and frequency tracking. Its corrections are filtered (low-pass with $\alpha = 0.2$) and rate-limited (maximum phase step 0.2 rad, maximum frequency step 0.05 Hz). Those safeguards are well matched to camera-derived flash timestamps, which carry inherent jitter from detection latency variability, frame-rate quantisation, and API communication delay.

This structural mismatch—continuous-phase coupling law operating on reconstructed phase from sparse events—is central to understanding the observed behavioural differences between the two models.

3.3 Formal dataset

The formal data used here are from the completed mutual visual HIL comparison run of 2026-06-21. The dataset path is recorded in the reproducibility appendix (Appendix D). The dataset includes:

- `aggregate_metrics.csv`
- `summary_by_model_condition.csv`
- `batch_metadata.json`
- `run_metadata.json`
- `chunk_metadata.json`

- per-trial `metrics_summary.json`, oscillator logs, detection logs, API logs, and flash-event logs

3.4 Models and locked parameters

The locked model parameters were read from the formal metadata and are shown in Table 1. EAPF Consensus uses the event-based phase/frequency consensus gains and limits selected in the earlier simulation-based model selection. Kuramoto uses the locked mutual-comparison gain $K = 5.0$.

Table 1: Locked model parameters used in the formal mutual HIL comparison.

Model	Parameters
EAPF Consensus	$g_p = 0.02$, $g_f = 0.02$, $\alpha_p = \alpha_f = 0.2$, $\Delta\theta_{\max} = 0.2$ rad, $\Delta f_{\max} = 0.05$ Hz
Kuramoto	$K = 5.0$

3.5 Trial schedule and seed control

The formal run was executed as three chunks, one per Pi initial frequency. The chunk metadata records all three completed chunks (Table 2). Each chunk completed six trials: two models by three repeats. Across all chunks this gives 18 completed trials and zero failed trials.

Random initial phases were enabled. The base seed was 20260620. Trial seeds were assigned using a canonical schedule over Pi frequencies 1.2, 1.5, and 2.5 Hz, so chunking did not change the phase schedule.

Table 2: Formal chunk validation records from chunk metadata.

Chunk	Freq pairs	Completed	Failed	Started
pi_1p2	2:1.2	6	0	2026-06-21T19:41:38.144053
pi_1p5	2:1.5	6	0	2026-06-21T19:48:33.342194
pi_2p5	2:2.5	6	0	2026-06-21T19:54:48.310484

3.6 Metrics

The core performance metrics are:

- actual virtual flash count ratio (FCR), using actual virtual flashes as the denominator;
- final 5 s mean absolute frequency error between Pi and virtual agents;
- absolute difference between the final 5 s mean Pi and virtual frequencies;
- final 5 s mean virtual and Pi frequencies;
- final-window frequency variability across each trial;
- Pi flash count, detected virtual flash count, and actual virtual flash count.

Runtime diagnostics were derived from per-trial logs. Capture FPS is estimated from detection-log sample count divided by trial duration. Processing FPS is the reciprocal of the mean camera processing time. API POST latency is measured from `api_events.csv`.

4 Results

4.1 Dataset validation

All expected formal trials were present: two models, three Pi initial frequencies, and three repeats. Each trial had a per-trial `metrics_summary.json`; no formal trial was excluded for corruption or incompleteness. The full validation table is included in Appendix A.

Table 3: Mutual HIL overall results across all formal trials (1 virtual + 1 Pi, 30 s, 3 repeats).

Model	Trials	Actual FCR	5 s MAE	Mean diff.	Virt. std	Pi std
EAPF Consensus	9	0.946 ± 0.028	0.046 ± 0.038	0.043	0.001	0.011
Kuramoto	9	0.903 ± 0.071	0.413 ± 0.095	0.199	0.247	0.393

Table 3 shows the main cross-condition result. EAPF Consensus has a mean final 5 s frequency error of 0.046 Hz, compared with 0.413 Hz for Kuramoto (locked $K = 5.0$). The final-window frequency variability is also much lower for EAPF: mean virtual frequency standard deviation is 0.001 Hz for EAPF and 0.247 Hz for Kuramoto. The Pi-side final-window standard deviation shows the same pattern (0.011 Hz vs. 0.393 Hz).

Table 4: Mean performance by model and Pi initial frequency condition. Frequency metrics are in Hz.

Pi	Model	Trials	FCR	5 s MAE	Mean diff.	Virt. final	Pi final
1.2 Hz	EAPF	3	0.960 ± 0.025	0.029 ± 0.019	0.029	1.908	1.879
1.2 Hz	Kuramoto	3	0.938 ± 0.057	0.440 ± 0.084	0.371	1.930	1.560
1.5 Hz	EAPF	3	0.937 ± 0.043	0.084 ± 0.045	0.084	1.949	1.866
1.5 Hz	Kuramoto	3	0.946 ± 0.037	0.348 ± 0.133	0.119	1.848	1.745
2.5 Hz	EAPF	3	0.941 ± 0.010	0.026 ± 0.015	0.015	2.126	2.139
2.5 Hz	Kuramoto	3	0.824 ± 0.043	0.451 ± 0.040	0.107	2.193	2.086

4.2 Robustness and detection

Figure 1 summarises two key axes: virtual flash detection reliability and final 5 s frequency agreement. Bars show the mean across three repeats per model-condition pair; black points show individual trial values. Both models generally retain usable virtual flash detection (most trials above the indicative FCR threshold of 0.85), but EAPF is more consistent. The difference is larger for frequency agreement, where Kuramoto has much higher error across all Pi initial frequency conditions.

4.3 Frequency agreement

Figure 2 shows the final 5 s mean frequencies by model and condition. EAPF keeps both virtual and Pi final frequencies close to the 2.0 Hz reference. Kuramoto final frequencies are more dispersed and condition-dependent. Figure 3 confirms this in scatter form: EAPF points cluster closer to the $y = x$ agreement line.

4.4 Representative trajectories

Figure 4 compares representative high-mismatch trials under the Pi 2.5 Hz initial condition (repeat 3). EAPF shows comparatively smooth virtual frequency adaptation. Kuramoto exhibits visibly larger oscillation in both the virtual and Pi frequency traces. This matches the observed frontend behaviour: the frontend in feedback-ON mode displays server `flash_on`, which is generated from server-side oscillator phase crossings.

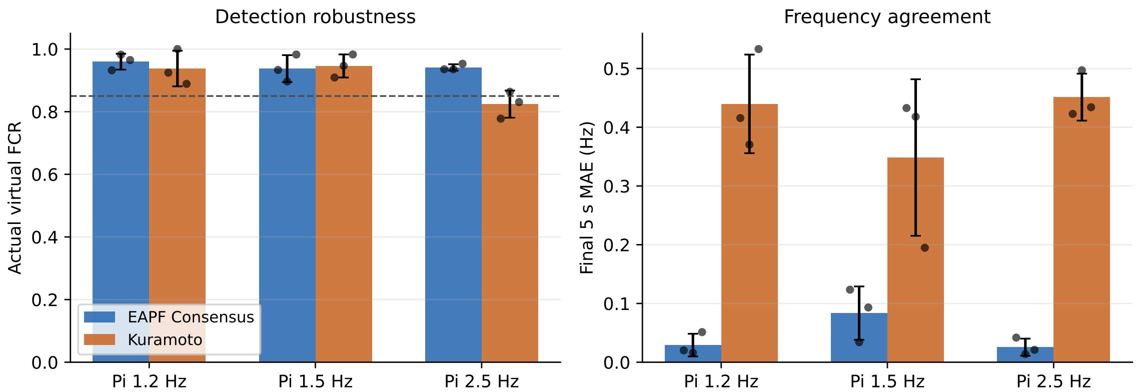
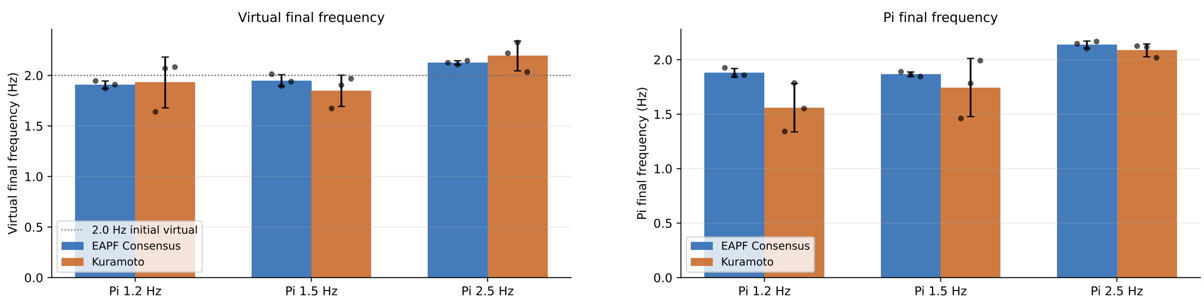


Figure 1: Mutual HIL robustness overview (1 virtual + 1 Pi, 30 s trials). Bars show mean across three repeats per model-condition pair; black points show individual trial values. The dashed line in the left panel marks an indicative actual-FCR threshold of 0.85.



(a) Virtual final frequency.

(b) Pi final frequency.

Figure 2: Final 5 s mean frequencies by model and condition. Bars show model-condition means; black points are individual trials.

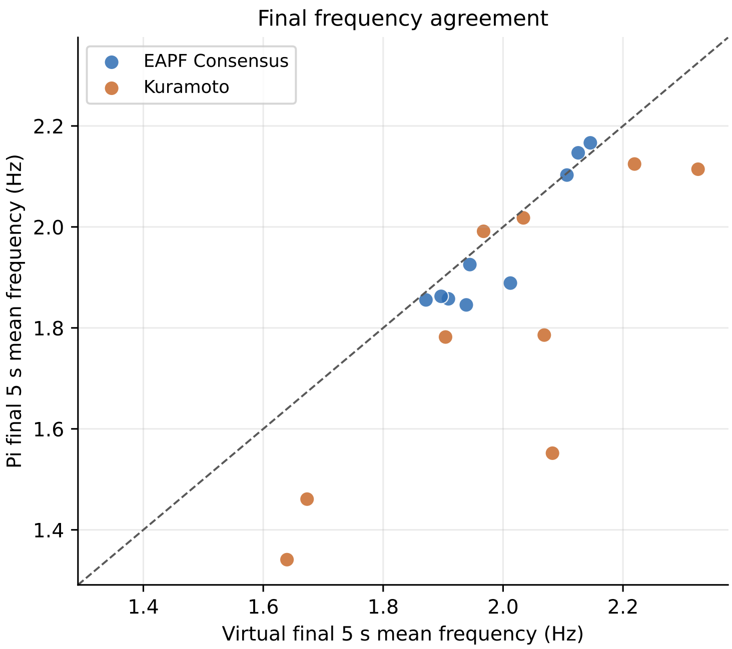


Figure 3: Final frequency agreement. Each point is one trial. The dashed line is the $y = x$ reference. EAPF points cluster closer to the agreement line.

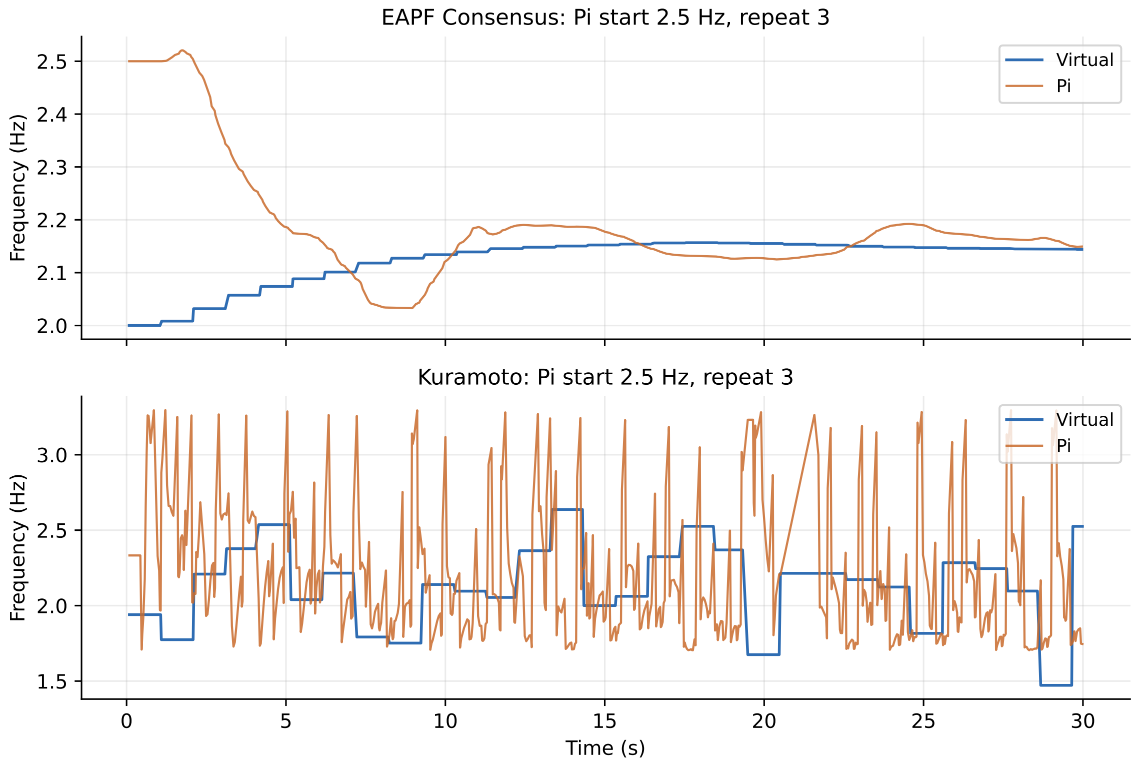


Figure 4: Representative per-trial frequency trajectories for the Pi 2.5 Hz initial condition, repeat 3 (30 s trials). The virtual frequency trace comes from `/api/agents`; the Pi trace comes from the local oscillator log.

4.5 Flash counts and runtime diagnostics

Figure 5 shows the flash-count diagnostics. Counts are broadly consistent with the 30 s trial duration and observed final frequencies. EAPF has slightly higher detected virtual flash counts overall, while Kuramoto shows lower actual FCR in the Pi 2.5 Hz condition.

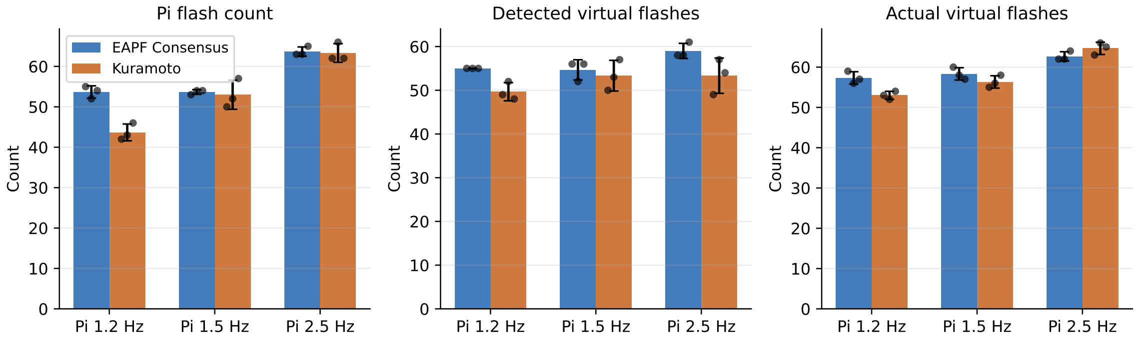


Figure 5: Pi flash count, detected virtual flash count, and actual virtual flash count by model and condition. Bars show means; points show individual trials.

Runtime diagnostics in Table 5 do not show a model-specific runtime failure. API success ratio was 1.000 for both models. Capture and processing rates were similar. This supports the interpretation that the main model difference is not an API or camera throughput artifact.

Table 5: Runtime and detection diagnostics derived from per-trial logs.

Model	**Cap. FPS**	**Proc. FPS**	**POST ms**	**API OK**	**Loop Hz**	**Events**
EAPF Consensus	21.8 ± 1.0	26.3 ± 0.8	18.1 ± 7.7	1.000	29.99	56.2
Kuramoto	22.1 ± 0.8	26.6 ± 0.9	21.9 ± 10.5	1.000	29.97	52.1

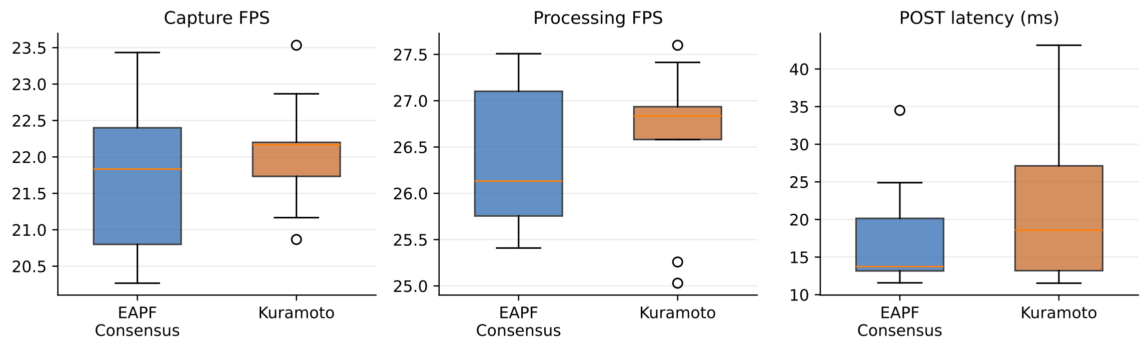


Figure 6: Runtime diagnostics derived from detection and API logs. Boxes show per-trial distributions across the 9 trials per model.

4.6 Initial phase sensitivity

The formal batch used random initial phases with controlled seeds. Figure 7 plots initial phase difference against final 5 s frequency error. With only three repeats per condition, this is an exploratory diagnostic rather than a formal phase-sensitivity study. It does show that Kuramoto's larger errors are not confined to a single initial phase offset.

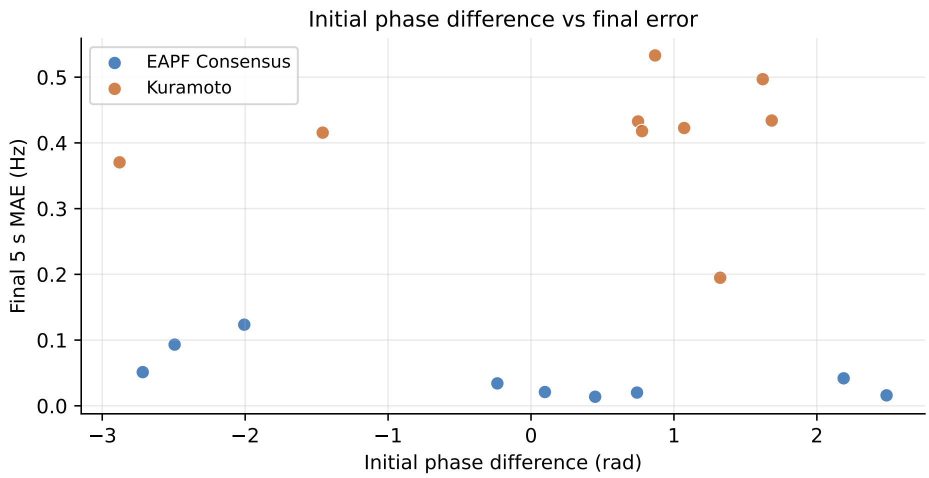


Figure 7: Initial phase difference versus final 5 s mean absolute frequency error. Each point is one trial.

5 Discussion

5.1 EAPF versus Kuramoto in mutual HIL

The formal mutual visual HIL data indicate that EAPF Consensus is more stable in the current setup. With locked $K = 5.0$, its final-window frequency errors are lower, its final-window frequency variability is much lower, and its behaviour is consistent across the three Pi initial frequencies. Kuramoto remains a useful comparison baseline, but in this mutual loop it shows larger fluctuations and poorer final-window agreement.

This conclusion is specific to the implemented event-based visual HIL system. It should not be read as a universal statement that EAPF is always superior to Kuramoto. Rather, EAPF appears better matched to this project's sensing and feedback pathway: timestamped flash events, camera-mediated detections, and explicit event-based phase/frequency correction with rate-limiting and filtering.

5.2 Why Kuramoto behaviour differs from fixed-leader results

The mutual HIL result does not contradict earlier fixed-leader visual results. Fixed-leader HIL is a one-way tracking task: the virtual leader flashes at a stable, externally controlled frequency and does not react to the Pi. Feedback perturbations affect only the follower, and cannot be amplified through the visual loop. In mutual HIL, both agents influence the timing that the other observes, forming a bidirectional closed loop.

Kuramoto can track a stable fixed visual reference well because the reference phase is not perturbed by the follower's own output. In mutual HIL, however, the virtual agent's phase is updated from Pi flash events, so the virtual visual stimulus itself carries the oscillations introduced by the Kuramoto phase-velocity correction. This feedback path does not exist in fixed-leader mode, which explains why a model can perform well in one-way tracking yet become less stable in mutual HIL.

The frontend instability observed during Kuramoto trials is consistent with this interpretation. The virtual Kuramoto update uses a direct instantaneous phase-velocity correction $\dot{\theta} = \omega + K\sin(\hat{\theta}_j - \theta)$. At the locked value $K = 5.0$, that correction can produce substantial short-timescale frequency swings. The frontend then displays the server's `flash_on` state, so visible flashing inherits those phase-crossing fluctuations.

EAPF avoids much of this behaviour because its corrections are filtered (low-pass $\alpha = 0.2$) and rate-limited ($\Delta\theta_{\max} = 0.2$ rad, $\Delta f_{\max} = 0.05$ Hz). This design is naturally better suited to the event-derived, latency-affected phase information available from camera flash detection.

5.3 Limitations

The formal sample size is modest: three repeats per model-condition pair (18 trials total). The interaction is camera-mediated and uses a screen-rendered target, which is not identical to drone-mounted LED-to-camera interaction. The Pi and frontend communicate through an HTTP API, and the browser display uses a fixed flash hold after server-side phase crossings. Kuramoto's gain was inherited from the earlier simulation comparison setting ($K = 5.0$) and may not be optimal for mutual visual HIL. A gain sensitivity analysis is presented in Appendix B.

6 Conclusion

The formal mutual visual HIL comparison (1 virtual + 1 Pi, 30 s trials, locked $K = 5.0$ for Kuramoto) supports carrying EAPF Consensus forward to the next 2-virtual + 1-Pi mutual HIL stage. EAPF achieved lower final-window frequency error and substantially lower final-window frequency variability than Kuramoto across all tested Pi initial frequencies.

Kuramoto is retained as a meaningful baseline. An appendix Kuramoto gain sensitivity sweep (Appendix B) tested $K \in \{2.5, 3.0, 3.5, 4.0, 4.5\}$ and found that $K = 2.5$ yields the best behaviour according to the appendix ranking score. However, even at $K = 2.5$, Kuramoto does not achieve the final-window stability of EAPF (see Appendix C). This supports the interpretation that the limitation is not merely an overly large coupling gain, but also the adaptation of a continuous-phase coupling law to sparse, event-derived phase estimates from camera flash detection.

The next experimental direction is therefore an EAPF-based 2-virtual + 1-Pi mutual HIL test, which will extend the mutual loop to three agents in a mixed-reality configuration.

A Trial Validation

Table 6: Per-trial validation for the formal mutual HIL dataset.

Trial	**Model**	**Pi Hz**	**Rep.**	**Aggregate**	**Metrics**	**Status**
eapf_consensus_V2_P1.2_r01	EAPF	1.2	1	yes	yes	included
kuramoto_V2_P1.2_r01	Kuramoto	1.2	1	yes	yes	included
eapf_consensus_V2_P1.5_r01	EAPF	1.5	1	yes	yes	included
kuramoto_V2_P1.5_r01	Kuramoto	1.5	1	yes	yes	included
eapf_consensus_V2_P2.5_r01	EAPF	2.5	1	yes	yes	included
kuramoto_V2_P2.5_r01	Kuramoto	2.5	1	yes	yes	included
eapf_consensus_V2_P1.2_r02	EAPF	1.2	2	yes	yes	included
kuramoto_V2_P1.2_r02	Kuramoto	1.2	2	yes	yes	included
eapf_consensus_V2_P1.5_r02	EAPF	1.5	2	yes	yes	included
kuramoto_V2_P1.5_r02	Kuramoto	1.5	2	yes	yes	included
eapf_consensus_V2_P2.5_r02	EAPF	2.5	2	yes	yes	included
kuramoto_V2_P2.5_r02	Kuramoto	2.5	2	yes	yes	included
eapf_consensus_V2_P1.2_r03	EAPF	1.2	3	yes	yes	included
kuramoto_V2_P1.2_r03	Kuramoto	1.2	3	yes	yes	included
eapf_consensus_V2_P1.5_r03	EAPF	1.5	3	yes	yes	included
kuramoto_V2_P1.5_r03	Kuramoto	1.5	3	yes	yes	included
eapf_consensus_V2_P2.5_r03	EAPF	2.5	3	yes	yes	included
kuramoto_V2_P2.5_r03	Kuramoto	2.5	3	yes	yes	included

B Kuramoto Gain Sensitivity

B.1 Motivation

In the main mutual visual HIL comparison (Section 4), Kuramoto with locked $K = 5.0$ showed visible frequency fluctuation and higher final-window variability than EAPF. This appendix tests whether lower Kuramoto coupling gains reduce the oscillation.

B.2 Theory framing

For a frequency mismatch Δf, the angular mismatch is $\Delta\omega = 2\pi\,\Delta f$. A one-way driven intuition gives a critical coupling $K_c \approx \Delta\omega$ for frequency locking. A symmetric two-oscillator mutual Kuramoto intuition gives $K_c \approx \Delta\omega/2$, because each oscillator carries half the coupling load.

This HIL system is not an ideal continuous Kuramoto system, because neighbour phase is reconstructed from binary flash events rather than observed continuously. The tested K values are therefore theory-informed diagnostics, not exact theoretical thresholds.

B.3 Experimental design

- virtual initial frequency = 2.0 Hz
- Pi initial frequencies = 1.2, 1.5, 2.5 Hz (matched to main comparison)
- K values = 2.5, 3.0, 3.5, 4.0, 4.5
- repeats = 2 per condition
- duration = 30 s

- random phases enabled; phase schedule paired across K values
- chunked execution; three chunks of 10 trials each; zero failed trials

The main locked-parameter Kuramoto comparison uses $K = 5.0$ and is not replaced by this appendix. This sweep is a secondary sensitivity analysis.

B.4 Results

Table 7: Kuramoto K-sweep: overall results across all Pi initial frequencies (2 repeats per condition).

K	Trials	FCR	5 s MAE	Mean diff.	Virt. std	Pi std	Score
2.5	6	0.844	0.432	0.392	0.213	0.257	0.847
3.0	6	0.864	0.320	0.261	0.198	0.277	1.730
3.5	6	0.873	0.362	0.249	0.262	0.286	2.498
4.0	6	0.845	0.399	0.213	0.273	0.368	2.703
4.5	6	0.868	0.332	0.123	0.179	0.344	2.461

Table 7 summarises results across all Pi initial frequencies. $K = 2.5$ achieves the best (lowest) appendix ranking score of 0.847. The ranking score penalises final-window mean absolute error, frequency variability, detection-FCR shortfalls, and diagnostic warning counts. Lower K values generally produce fewer Kuramoto debug warnings (mean 1.8 at $K = 2.5$ vs. 87.8 at $K = 4.5$), consistent with reduced aggressive phase-velocity correction.

Table 8: Kuramoto K-sweep results by Pi initial frequency condition.

Pi	K	Trials	FCR	5 s MAE	Virt. std	Pi std	Score
1.2 Hz	2.5	2	0.877	0.687	0.263	0.251	1.029
1.5 Hz	2.5	2	0.935	0.315	0.196	0.258	0.632
2.5 Hz	2.5	2	0.721	0.293	0.180	0.261	0.881
1.2 Hz	3.0	2	0.952	0.560	0.355	0.324	2.140
1.5 Hz	3.0	2	0.900	0.213	0.155	0.272	1.103
2.5 Hz	3.0	2	0.740	0.186	0.086	0.234	1.946
1.2 Hz	3.5	2	0.932	0.618	0.464	0.360	2.819
1.5 Hz	3.5	2	0.918	0.151	0.093	0.158	0.977
2.5 Hz	3.5	2	0.767	0.318	0.229	0.340	3.698
1.2 Hz	4.0	2	0.934	0.476	0.350	0.362	2.433
1.5 Hz	4.0	2	0.844	0.370	0.291	0.371	2.104
2.5 Hz	4.0	2	0.757	0.352	0.178	0.370	3.572
1.2 Hz	4.5	2	0.931	0.315	0.063	0.389	1.911
1.5 Hz	4.5	2	0.938	0.376	0.326	0.260	2.159
2.5 Hz	4.5	2	0.736	0.306	0.150	0.383	3.311

Table 9: Kuramoto K ranking from the appendix sensitivity sweep. Lower score is better.

Rank	K	Score	5 s MAE	Fan-out warnings
1	2.5	0.847	0.432	1.8
2	3.0	1.730	0.320	52.7
3	4.5	2.461	0.332	87.8
4	3.5	2.498	0.362	88.5
5	4.0	2.703	0.399	92.5

Table 8 breaks results down by Pi initial frequency. $K = 2.5$ is the best-performing gain in every condition. Table 9 shows the overall ranking.

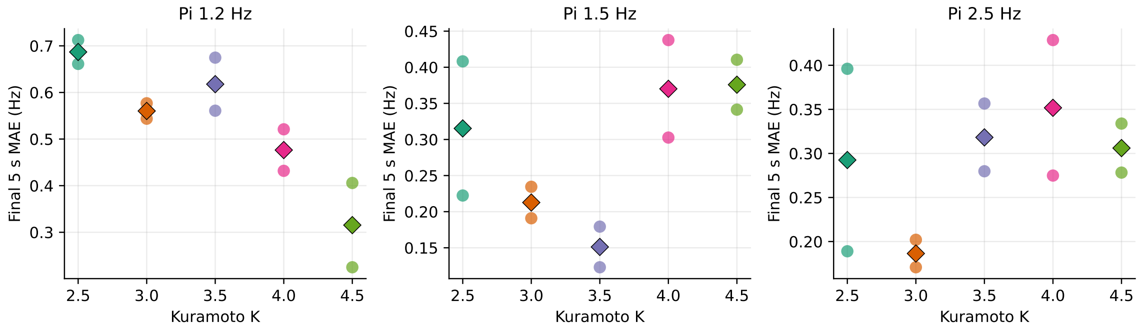


Figure 8: Final 5 s mean absolute frequency error vs. Kuramoto K, by Pi initial frequency. Filled diamonds show the mean across 2 repeats; open circles show individual trial values.

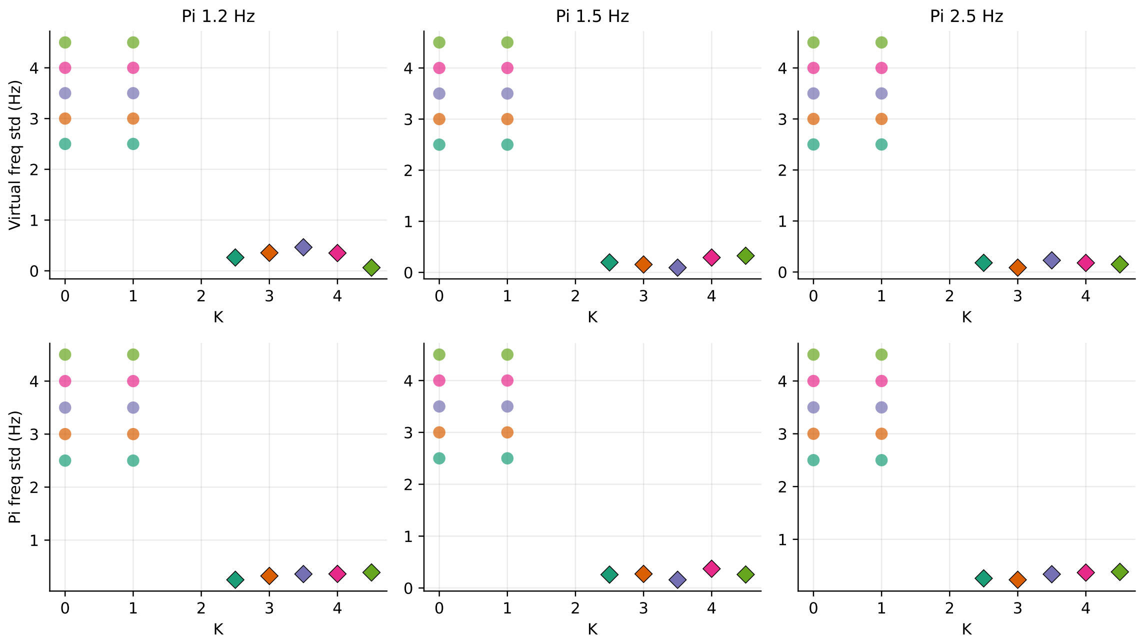


Figure 9: Final 5 s frequency standard deviation vs. K. Top row: virtual frequency std; bottom row: Pi frequency std.

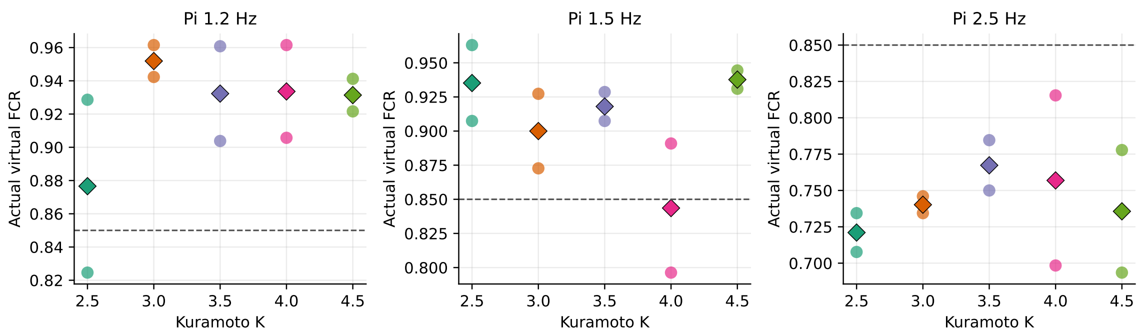


Figure 10: Actual virtual flash count ratio vs. K. The dashed line marks FCR = 0.85.

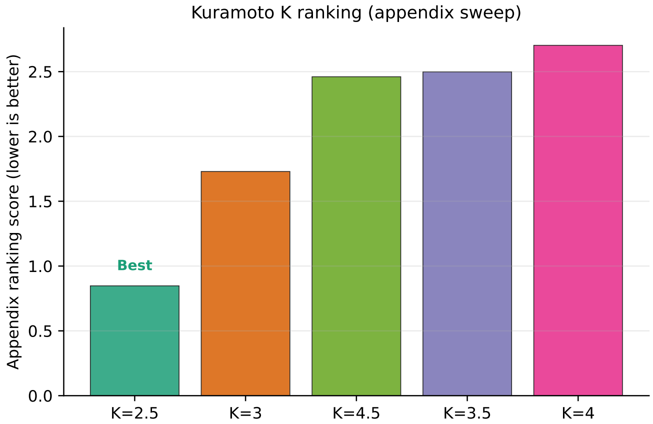


Figure 11: Appendix ranking score vs. K (lower is better). $K = 2.5$ is best. Score formula: score $= \mathrm{MAE} + 0.5(\sigma_{\mathrm{virt}} + \sigma_{\mathrm{Pi}}) + 2 \cdot \max(0, 0.90 - \mathrm{FCR}) + 5 \cdot \max(0, \mathrm{FCR} - 1.02) + 0.02 \cdot (\mathrm{warnings})$, with zero failed trials.

B.5 Interpretation

Lower K reduces the aggressive instantaneous phase-velocity correction compared with $K = 5.0$. $K = 2.5$ is the best tested value according to the appendix ranking. However, the sweep shows that Kuramoto does not fully converge to smooth, low-variability behaviour in the mutual visual HIL loop even at this best gain. The remaining oscillation suggests that the issue is not only gain magnitude, but also the mismatch between continuous-phase Kuramoto coupling and event-based visual flash timing.

Reducing the Kuramoto gain improves the behaviour relative to the locked $K = 5.0$ setting, but does not fully remove the oscillatory response. This supports the interpretation that the limitation is not merely an overly large coupling gain, but also the adaptation of a continuous-phase coupling law to sparse camera-derived flash events.

C Best tested Kuramoto gain versus EAPF

This section compares the existing main EAPF data with the best tested Kuramoto gain $K = 2.5$ from the appendix sweep. No new trials were run; all data are from the completed formal datasets.

The comparison should be interpreted with care:

- EAPF has 3 repeats per condition in the main dataset.
- Kuramoto $K = 2.5$ has 2 repeats per condition in the appendix sweep.
- The phase schedules are controlled and paired, but sample sizes differ.
- This is a secondary appendix comparison, not the main locked-parameter result.

Table 10 compares the two models across conditions. EAPF maintains lower final-window frequency variability (virtual std and Pi std) in all conditions. The 5 s MAE values are more comparable at the moderate-mismatch Pi 1.5 Hz condition, but EAPF is consistently lower at the higher-mismatch Pi 1.2 Hz and Pi 2.5 Hz conditions.

C.1 Interpretation

$K = 2.5$ improves Kuramoto relative to $K = 5.0$ on several metrics, especially in reducing the excessive warning/oscillation score. However, EAPF still has much lower final-window frequency

Table 10: EAPF Consensus (main) versus best-appendix Kuramoto K=2.5. Values shown as "EAPF / K=2.5". Sample sizes differ: EAPF 3 repeats, K=2.5 has 2 repeats.

Pi	N (E/K)	5 s MAE (Hz)	Virt. std (Hz)	Pi std (Hz)	Actual FCR
1.2 Hz	3	2	0.029 / 0.687	0.000 / 0.263	0.008 / 0.251
0.960 / 0.877					
1.5 Hz	3	2	0.084 / 0.315	0.001 / 0.196	0.003 / 0.258
0.937 / 0.935					
2.5 Hz	3	2	0.026 / 0.293	0.001 / 0.180	0.022 / 0.261
0.941 / 0.721					

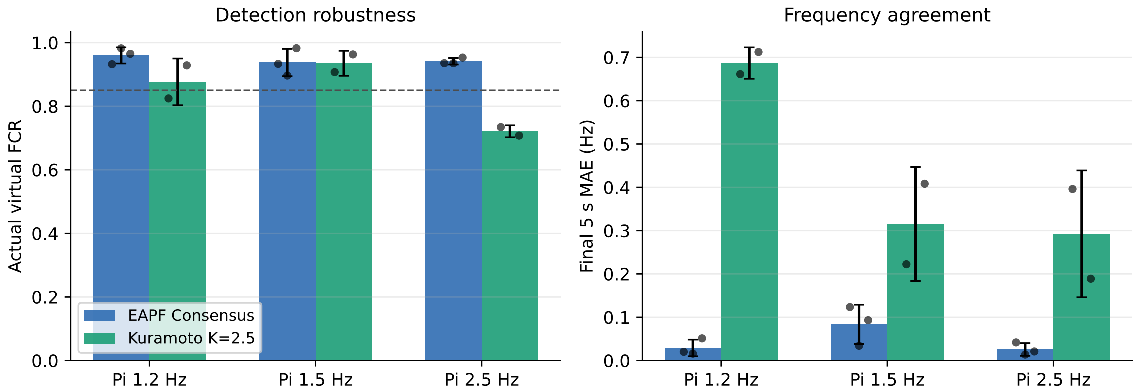


Figure 12: EAPF Consensus (main, 3 repeats) vs. Kuramoto $K = 2.5$ (appendix, 2 repeats). Left: actual virtual FCR; right: final 5 s MAE. Bars show means; points show individual trials.

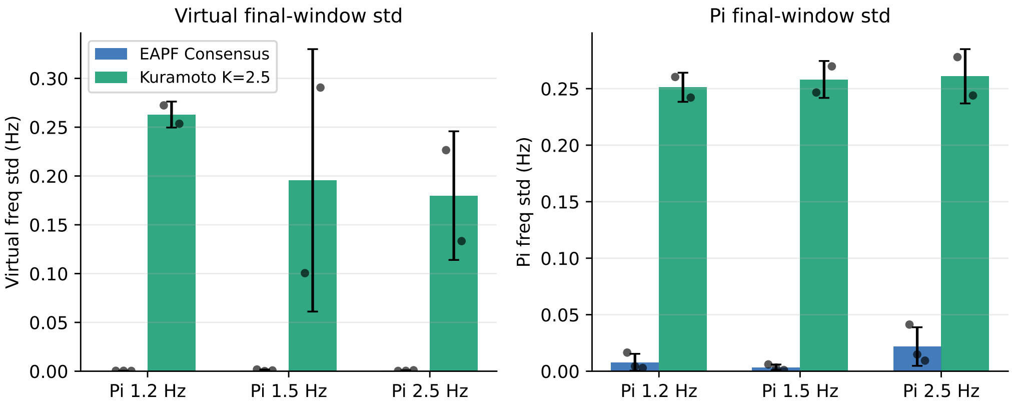


Figure 13: Final-window frequency standard deviation: EAPF vs. Kuramoto $K = 2.5$. Left: virtual frequency std; right: Pi frequency std.

variability and better stability across conditions.

Therefore the model selection for the next 2-virtual + 1-Pi mutual HIL test remains EAPF Consensus. This conclusion does not claim that EAPF universally outperforms Kuramoto in all possible implementations. It states that EAPF is better suited to this project's current binary-flash, camera-mediated, event-based mutual HIL setup.

D Reproducibility

The main formal dataset (1 virtual + 1 Pi mutual HIL comparison) is stored at:

```
experiments/logs/step5b_formal_chunked/formal_step5b_chunked_20260621
```

The Kuramoto K-sensitivity appendix dataset is stored at:

```
experiments/logs/step5b_kuramoto_k_sensitivity/k_sweep_20260621_v4
```

The figures and tables in this report were generated by:

```
docs/mutual_visual_hil_report/generate_mutual_hil_analysis.py
```

The directory names `step5b_formal_chunked` and `step5b_kuramoto_k_sensitivity` in the experiment log paths are internal folder labels only. The conceptual naming used throughout this report is "mutual visual HIL" and "Kuramoto gain sensitivity appendix."

Derived copies of the core CSV/JSON inputs and intermediate summary tables are stored under:

```
docs/mutual_visual_hil_report/derived/
```